

Maximize or minimize

$$1 = \max_{x \in \mathbb{R}} -x^2 + 1 \text{ s.t. } x \in \mathbb{R}$$

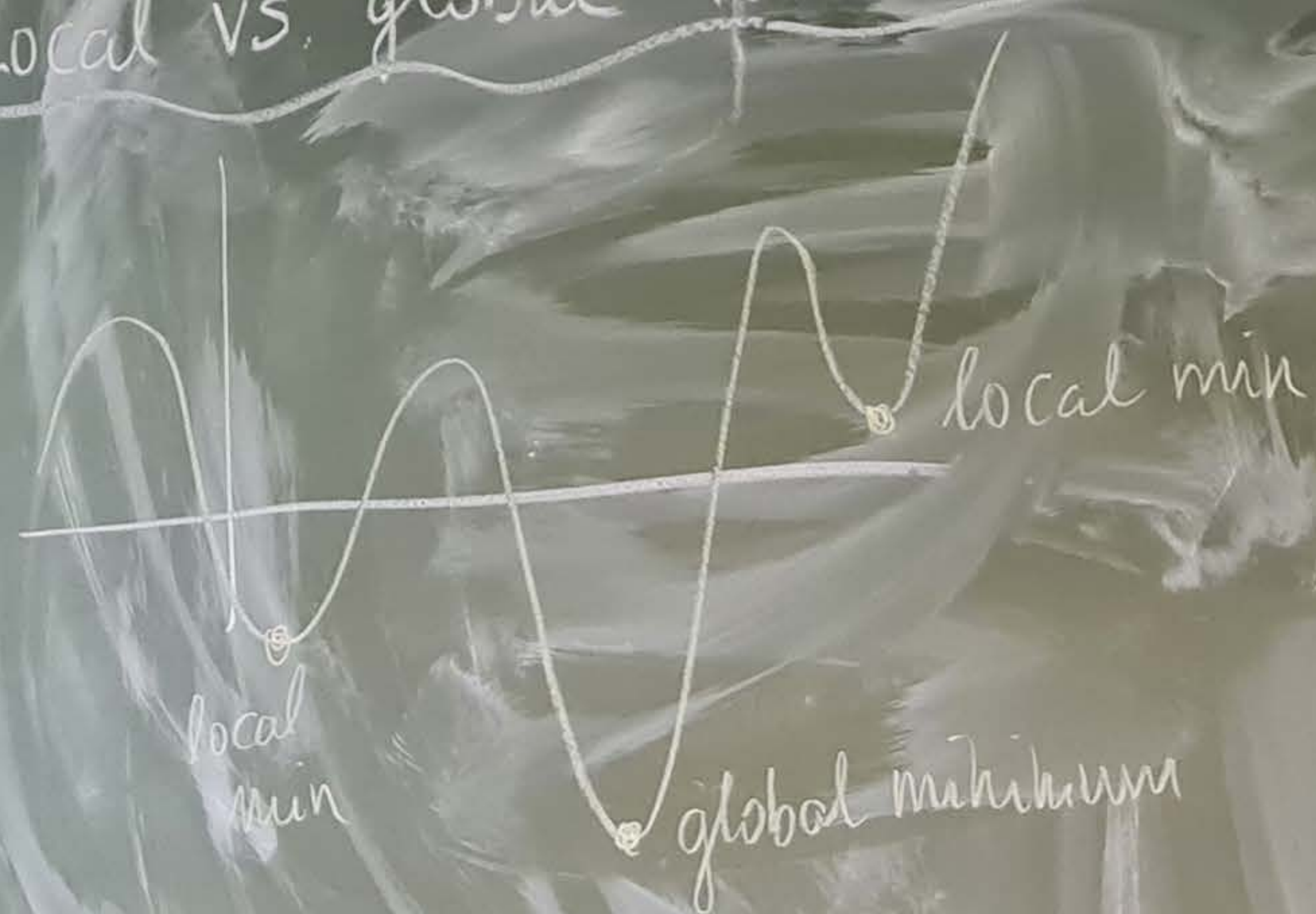
$$x_* = 0$$

$$-1 = \min_{x \in \mathbb{R}} \underbrace{-(-x^2 + 1)}_{x^2 - 1} \text{ s.t. } x \in \mathbb{R}$$

$$x_* = 0$$



Local vs. global minima

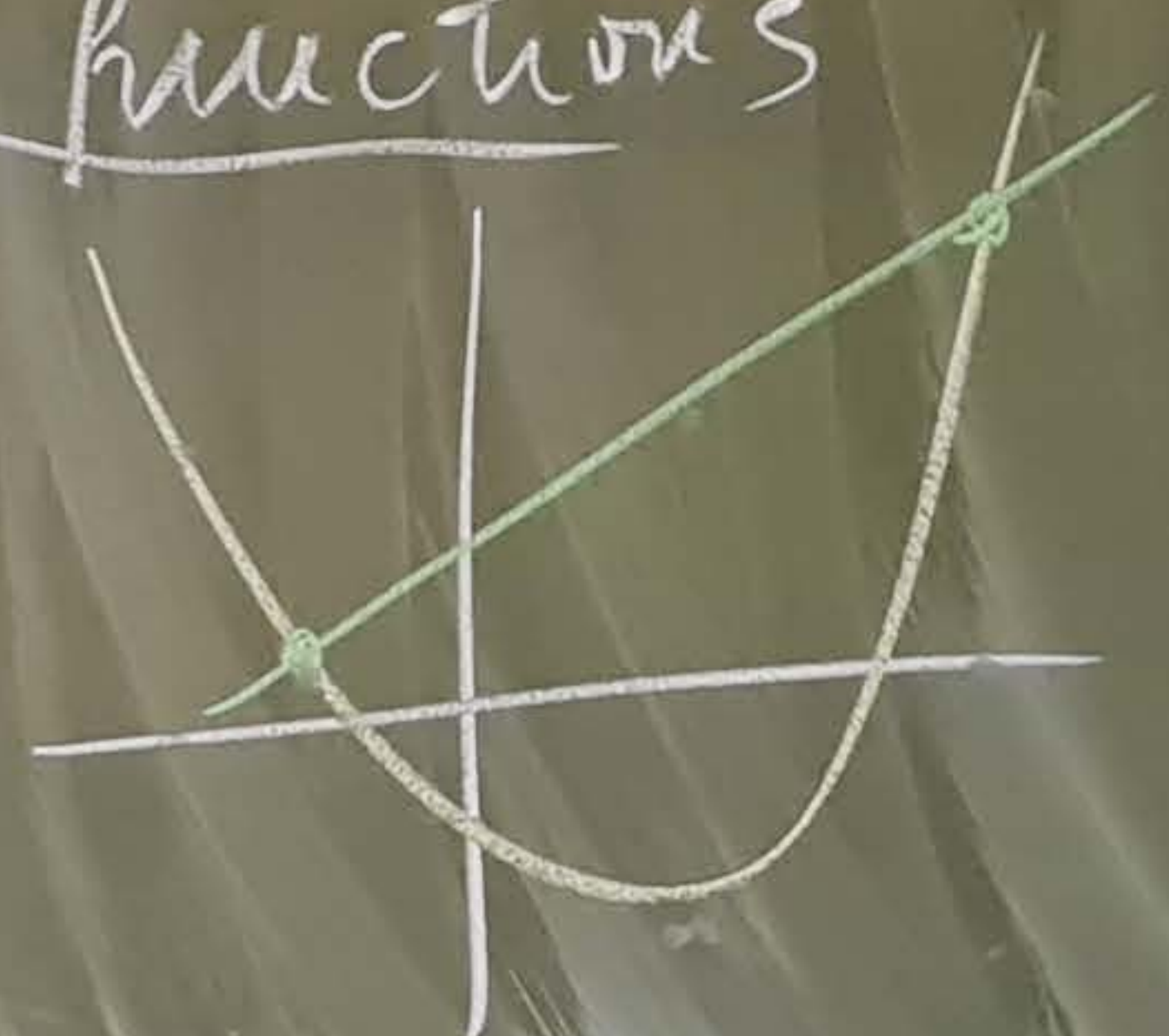


Example $f(x, y) = x^2 + 10y^2$
 $(x_*, y_*) = (0, 0)$

$$\text{grad } f(x, y) = \begin{pmatrix} 2x \\ 20y \end{pmatrix}$$

$$(x_0, y_0) = (2, 2)$$

Convex functions



Example $f(x, y) = x^2 + 10y^2$

1) $(x_0, y_0) = (2, 2) = \vec{x}_0$

2) $\nabla f(2, 2) = \begin{pmatrix} 4 \\ 40 \end{pmatrix} = \vec{d}_0$, $\|\nabla f(2, 2)\| = \sqrt{4^2 + 40^2} \approx 40$

3) $\alpha_0 = 1$

$$f(x_0, y_0) - 1 \cdot (4, 40) = f(2-4, 2-40) = f(-2, -38) = (-2)^2 + 10(-38)^2 \gg f(2, 2)$$

no descent in f

a smaller step size is needed

help function:

$$\varphi(\alpha) = f(\vec{x}_k - \alpha \vec{d}_k)$$

1st idea:

$$\alpha_k = \min_{\alpha} \varphi(\alpha)$$

2nd idea:

heuristic choice

quite computational intensive and it is not guaranteed that it is globally the best choice
 (Armijo-Goldstein-Rule)